

SPEAK

Optical Sensor Bridge

AFBR optical receiver + ESP32-S3 BLE + iPhone/iPad

PRODUCTION-TARGET DESIGN • SEPTEMBER 2026

STATUS: Architecture approved; implementation planned. Hardware validation is the next acceptance step.

Signal path

1 mm POF > Broadcom AFBR-S10RX0x1Z > protected ADC > ESP32-S3 > BLE GATT > Speak

Technical basis

- Receiver supply: 4.75-5.25 V (nominal 5 V); maximum specified VOUT: 4 V.
- Photosensitivity range: 300-1100 nm; maximum photosensitivity at 650 nm.
- Typical rise/fall: 40 ns / 60 ns. Analog scaling/protection precedes the ESP32-S3 ADC.

Fail-closed Speak controls

- Speak must be ON and the user must explicitly select the BLE sensor source.
- Schema, freshness, sequence, calibration, range and quality gates must all pass.
- Any failed gate or disconnect forces zero sensor effect; the app-level volume cap remains authoritative.
- The sensor cannot turn Speak on, open the microphone, raise the volume cap or trigger external actions.

Controlled bench pilot

- Connect AFBR + 1 mm POF + protected ESP32-S3 bridge.
- Stream versioned optical-level and quality telemetry into Speak on iPhone/iPad.
- Calibrate baseline/reference levels; challenge clipping, stale frames and disconnects.
- Produce an acceptance receipt containing firmware, schema, calibration and gate decisions.

Scope boundary

The AFBR receiver measures optical power. This integration does not label the channel as thought detection, neurological evidence, identity, diagnosis, mood, cognition or intent.

PILOT / ENGINEERING CONVERSATION

MVPuknowme/MVPuknowme -> second-hub/speak

Technical basis: Broadcom pub-005792. Independent integration; no Broadcom endorsement is stated or implied.